Does earnings information predict abnormal returns?

Building the pipeline so that the answer can be trusted — and proving it on data where the answer is known.

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ABSTRACT

Post-earnings-announcement drift is among the most-studied anomalies in finance and among the easiest to reproduce spuriously. This note describes an end-to-end research engine — point-in-time data acquisition, event alignment, abnormal-return measurement, purged walk-forward modelling, and a sector-neutral book net of costs — and validates it on a synthetic market whose data-generating process is known. Across six rolling annual holdouts the model attains a mean out-of-sample rank IC of 0.197 (positive in 6/6 years) and a net Sharpe of 4.82. Run identically on data with *no* effect planted, the same pipeline returns an IC of 0.022 and a net Sharpe of 0.37. That contrast, not the headline number, is the result: it establishes that the machinery detects a signal when one exists and does not manufacture one when it does not.

1 · DESIGN

For each year *Y*, the model is fitted only on announcements whose 20-day outcome had fully resolved before 1 January *Y*, less a 25-session embargo; it is then frozen and used to predict every announcement in *Y*. The target is the cumulative abnormal return from **one session after** the announcement to twenty sessions after. Windows beginning on the announcement day contain the opening gap, which cannot be traded, and booking it is the single largest source of overstated drift results. Abnormal returns are computed three ways — market-adjusted, market-model, and leave-one-out sector — because "abnormal" is a modelling choice and reporting one number hides how much of the result is that choice. Features are fundamental changes differenced year-on-year, standardised unexpected earnings, and Loughran–McDonald text measures with their period-on-period deltas; each carries the timestamp at which it became public, and a single guard refuses any panel where a feature post-dates the moment of trading.

2 · SIX HELD-OUT YEARS

Year	Train n	Test n	IC	IC t	Pred. bp	Real. bp	Calib.	Sharpe
2019	2,594	600	0.202	3.95	602	414	0.79	2.62
2020	3,194	600	0.211	6.91	561	586	0.92	4.35
2021	3,794	600	0.236	14.39	560	627	0.98	5.75
2022	4,394	600	0.164	2.68	559	445	0.71	5.01
2023	4,994	600	0.169	5.09	548	490	0.80	5.11
2024	5,594	600	0.198	9.43	550	435	0.87	5.65

Table 1. Each row is a year the model never saw in training. *Pred.* and *Real.* are the predicted and realised top-minus-bottom quintile spreads in basis points; *Calib.* is the slope of realised on predicted, where 1.00 would mean the predicted magnitudes were exactly right.

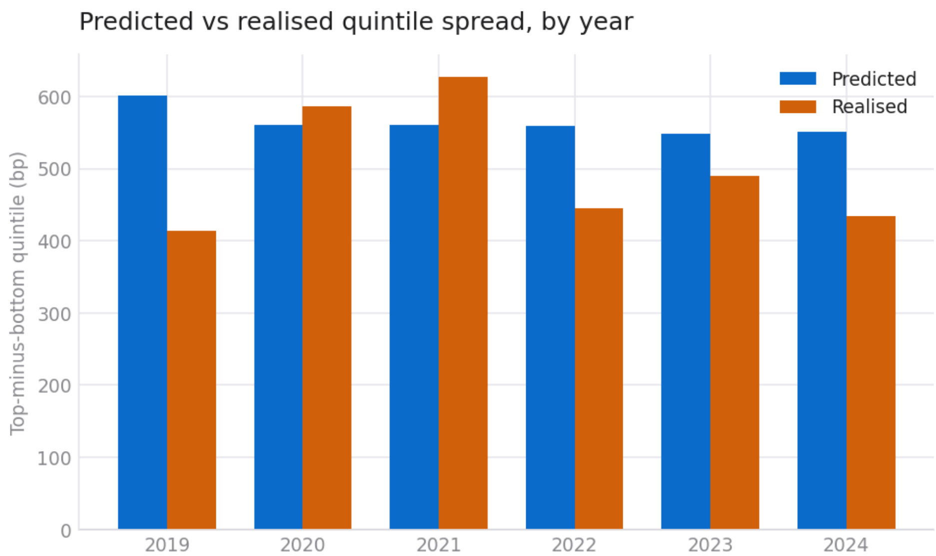

Figure 1. Predicted against realised quintile spread. The direction is right in all six years; the magnitude is optimistic in five.

3 · THE NULL CONTROL

The same pipeline was run on a market generated with no post-announcement effect at all. It still produces predictions, and still produces confident ones — a predicted spread near 200 bp in every year. In 2021 it posts an IC of 0.080 at t = 3.67 with a net Sharpe of 3.79: a textbook false positive, in data where there is provably nothing to find. Anyone reporting a single year would have reported that one. Across all six years the effect vanishes, which is the behaviour required of an honest pipeline.

Statistic	Effect planted	Null control
Mean out-of-sample IC	0.197	0.022
t across the six years	17.99	1.15
Years with positive IC	6/6	4/6
Mean realised spread (bp)	499	89
Mean calibration slope	0.85	0.27
Net Sharpe, years stitched	4.82	0.37
Newey–West t on daily P&L	8.27	0.70

Table 2. Identical code, identical seed, identical universe; the only difference is whether an effect exists in the data.

4 · IS IT ALPHA, AND IS IT NEW?

Regressed on Fama–French five factors plus momentum with Newey–West errors, the book shows an annualised alpha of 8.76% (t = 8.33) and an R^2 of 0.011 — no loading significant at 5%. The caveat is material: these are synthetic proxies built from the same panel, not Ken French data, so the regression is demonstrated rather than the neutrality. Every specification evaluated is logged — eight, including the two abandoned — and the deflated Sharpe is deliberately *not* reported, because only two of the eight carry a Sharpe and a dispersion estimated from two numbers would give a flattering hurdle. The code refuses and says so. A Fama–MacBeth regression over 83 monthly cross-sections attributes most of the effect to one fundamental feature (revenue growth acceleration, +35 bp per standard deviation, t = 2.47) and finds the text features individually insignificant — collinearity between three measures of one latent quantity, not a contradiction of the sort.

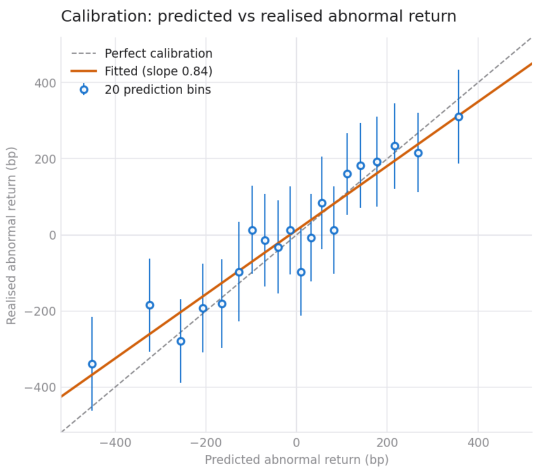

Figure 2. Predictions binned into twenties against realised outcomes. A fitted slope of 0.84 against a perfect 1.00: the ordering is informative, the magnitudes are overstated.

5 · WHAT THIS DOES NOT ESTABLISH

The market is synthetic and its planted drift is roughly three times anything documented in real equities — deliberately, so the demonstration clears the noise in a short sample. Nothing here is a claim about tradable returns. Six years is six observations. The generating process is linear, Gaussian and stationary, so recovering a linear effect says nothing about a non-linear one. Costs are assumed rather than measured, and no capacity analysis has been done. The null control is a single draw. On real data, the documented decay in this anomaly since roughly 2004 means an IC near 0.03 and a net Sharpe below 1 would be a good outcome — and a Sharpe of 4 would mean a bug.

6 · REPRODUCIBILITY

`make reproduce` regenerates every figure and number in this note and writes a SHA-256 for each of the 36 published artefacts; `make verify` re-runs and diffs against the committed manifest, and continuous integration fails if a hash moves. The full method, the pre-registered falsification criteria, and a catalogue of the eight ways this study could flatter itself are at github.com/pingadom/Earnings-Event-Research.